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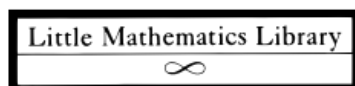


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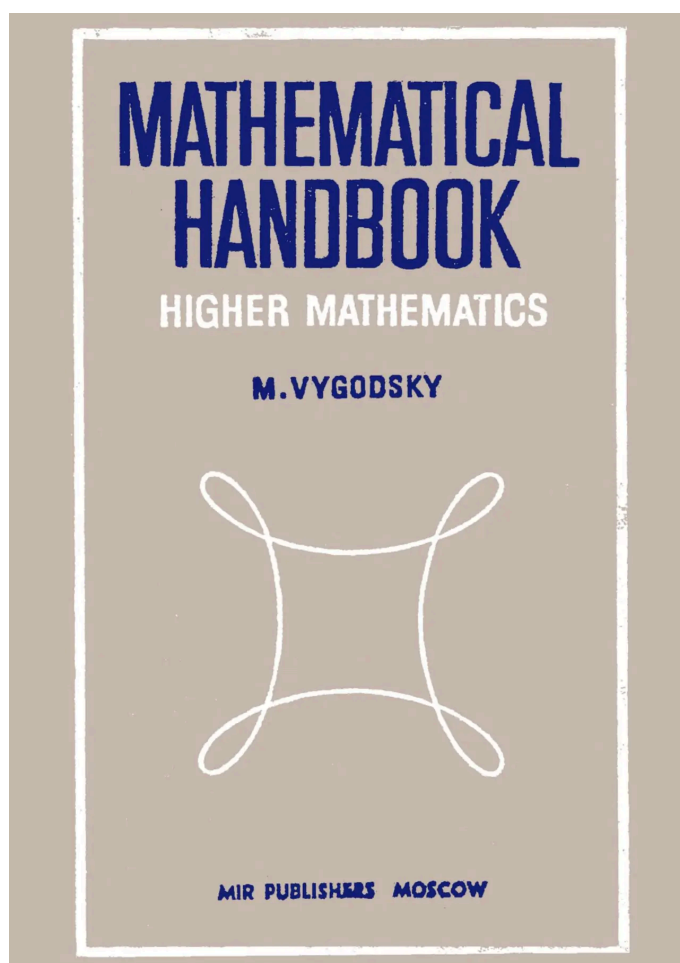
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Математический справочник — высшая математика — Выгодский

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В этом посте мы рассмотрим книгу «Математический справочник — высшая математика» М. Выгодского. Ранее мы уже рассматривали «Математический справочник — элементарная математика» того же автора.



О книге

Это пособие является продолжением «Справочника по элементарной математике» того же автора и включает в себя материал, который обычно изучается на математических курсах в высших учебных заведениях.

Назначение этого пособия двоякое.

Во-первых, это справочник, в котором читатель может найти определения (что такое векторное произведение?) и фактическую информацию, например о том, как найти поверхность тела вращения или как разложить функцию в тригонометрический ряд, и так далее. Определения, теоремы, правила и формулы (сопровожаемые примерами и практическими советами) легко найти по обширному указателю или оглавлению.

Во-вторых, пособие предназначено для систематического изучения. Оно не заменяет учебник, поэтому полные доказательства приводятся только в исключительных случаях. Тем не менее оно вполне может послужить материалом для первого знакомства с предметом. Для этого подробно объясняются основные понятия, такие как скалярное произведение (§ 104), предел (§ 203–206), дифференциал (§ 228–235) или бесконечные ряды (§ 270, 366–370). Все правила подробно проиллюстрированы примерами, которые являются неотъемлемой частью руководства (см. разделы 50–62, 134, 149, 264–266, 369, 422, 498 и другие). В пояснениях указано, как действовать, если правило перестает быть актуальным, а также перечислены ошибки, которых следует избегать (см. разделы 290, 339, 340, 379 и другие).

Теоремы и правила сопровождаются обширным пояснительным материалом. В некоторых случаях акцент делается на раскрытии содержания теоремы, чтобы облегчить понимание доказательства. В других случаях приводятся специальные примеры, а рассуждения строятся таким образом, чтобы обеспечить полное доказательство теоремы при применении к общему случаю (см. разделы 148, 149, 369, 374). Иногда в качестве пояснения читателю просто предлагается обратиться к разделам, на которых основано доказательство. Материал, напечатанный мелким шрифтом, можно пропустить при первом прочтении, однако это не значит, что он не важен.

Значительное внимание было уделено историческому контексту математических понятий, их происхождению и развитию. Это очень часто помогает пользователю взглянуть на предмет в правильном ракурсе. Особый интерес в этом отношении представляют разделы 270, 366, а также разделы 271, 383, 399 и 400, которые, как мы надеемся, дадут читателю

более четкое представление о рядах Тейлора, чем то, которое обычно можно получить при формальном изложении. Кроме того, там, где это было сочтено целесообразным, мы включили биографические сведения из жизни математиков.

Книга была переведена с русского языка Георгием Янковским и опубликована в 1987 году (пятое издание) издательством «Мир».

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PS: я подумываю о том, чтобы разместить это пособие (наряду с пособием по элементарной математике) и пособия по физике Детлафа и Яворского на специальном сайте, чтобы их можно было легко найти в интернете. Это также позволит создать постоянные URL-адреса для конкретных понятий, на которые можно будет легко ссылаться. Мы будем рады любым идеям о том, какую платформу/технология использовать. На данный момент я подумываю о создании сайта на чистом HTML с поддержкой MathML, но, пожалуйста, предложите любую другую платформу, которая больше подходит для этого проекта.

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